

Student Number _____

ASCHAM SCHOOL

**2022****YEAR 12****TRIAL****EXAMINATION**

Mathematics

Extension 2

General Instructions

- Reading time – 10 minutes.
- Working time – 3 hours.
- Write using black non-erasable pen.
- NESA-approved calculators may be used.
- A NESA Reference Sheet is provided.
- All necessary working should be shown in every question.

Total marks – 100

- Attempt Sections A and B.
- Section A is worth 10 marks.
- Recommended time on Section A: 15 minutes
- Answer Section A on the multiple choice answer sheet.
- Detach the multiple choice answer sheet from the back of the examination paper.
- Section B contains 6 questions worth 15 marks each.
- Recommended time on Section B: 2 hours 45 minutes
- Answer each question in a new booklet.
- Label all sections clearly with your name/number and teacher.

SECTION A – 10 MULTIPLE CHOICE QUESTIONS 10 MARKS**ANSWER ON THE ANSWER SHEET****1**

Which of the following vectors is perpendicular to the vector $\underline{u} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and the y-axis?

A $\underline{m} = \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix}$

B $\underline{m} = \begin{pmatrix} 0 \\ 3 \\ -2 \end{pmatrix}$

C $\underline{m} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$

D $\underline{m} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$

2

What is the minimum value of $|2x - y| + |x + 2y|$?

A $|x + 3y|$

B $|x - 3y|$

C $|3x - y|$

D $|3x + y|$

3 Which of the following is true?

A $\forall x, y \in \mathbb{N}, \exists a, b \in \mathbb{Q} \quad \frac{1}{x+iy} = a+ib$

B $\forall x, y \in \mathbb{R}, \exists a, b \in \mathbb{Z} \quad \frac{1}{x+iy} = a+ib$

C $\forall x, y \in \mathbb{Q}, \exists a, b \in \mathbb{N} \quad \frac{1}{x+iy} = a+ib$

D $\forall x, y \in \mathbb{Z}, \exists a, b \in \mathbb{Z} \quad \frac{1}{x+iy} = a+ib$

4 It is known that $1-2i$ is a root of $z^4 + az^3 + bz^2 + 10 = 0$, where a and b are real.
Which of the following could also be a root of $z^4 + az^3 + bz^2 + 10 = 0$?

A $1+i$

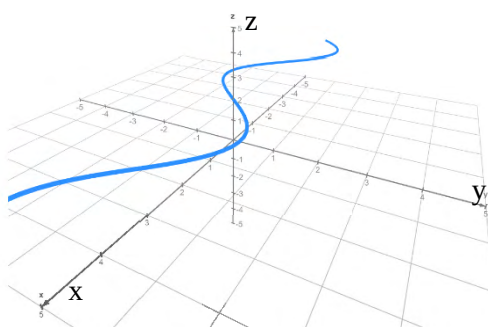
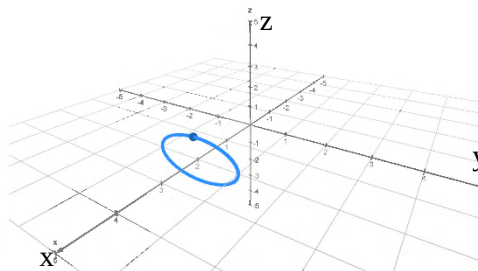
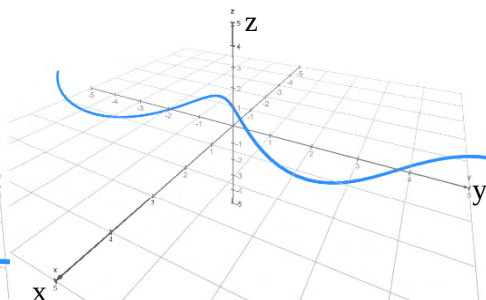
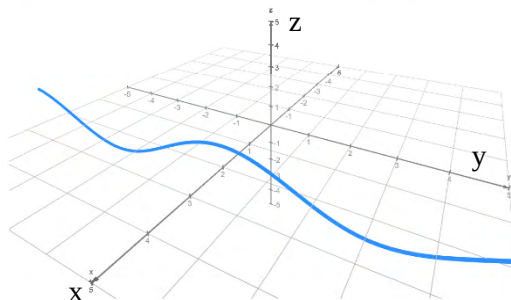
B 2

C $1+2i$

D All of the above

5

What is a possible graph for the parametric equation $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ t \\ \cos t \end{pmatrix}$?

A**B****D****C**

6 Which of the following is the contrapositive of the following statement?

$\forall x, y, z, n \in \mathbb{N} : \text{if } x^n + y^n \neq z^n \text{ then } n \geq 3.$

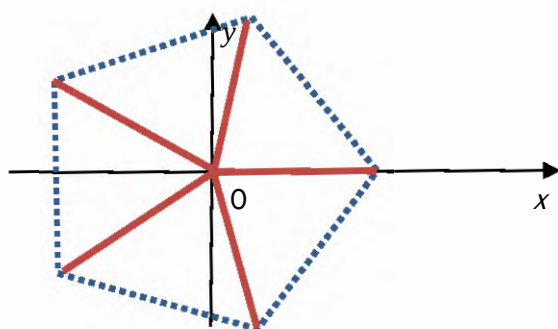
A $\forall x, y, z, n \in \mathbb{N} : \text{if } n \geq 3 \text{ then } x^n + y^n \neq z^n.$

B $\forall x, y, z, n \in \mathbb{N} : \text{if } n < 3 \text{ then } x^n + y^n \neq z^n.$

C $\forall x, y, z, n \in \mathbb{N} : \text{if } n \geq 3 \text{ then } x^n + y^n = z^n.$

D $\forall x, y, z, n \in \mathbb{N} : \text{if } n < 3 \text{ then } x^n + y^n = z^n.$

7 The graph below shows roots of unity on the Argand diagram.



What is the equation?

A $z^4 = -1$

B $z^5 = -1$

C $z^4 = 1$

D $z^5 = 1$

8 What is the value of $\sum_{1}^{99} i^n$?

A i

B -1

C $-i$

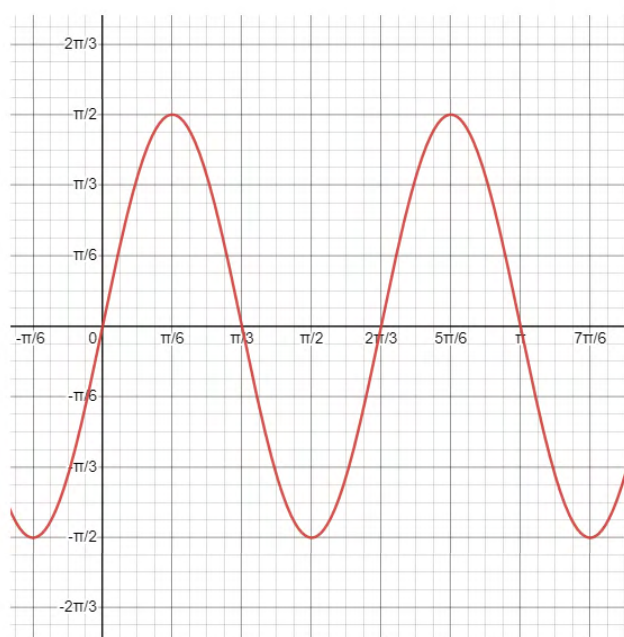
D 1

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- 9 Consider the complex numbers z such that $|z - 2i| = 1$. What is the maximum value of $\arg z$?

- A $\frac{\pi}{4}$
 B $\frac{\pi}{3}$
 C $\frac{2\pi}{3}$
 D $\frac{3\pi}{4}$

- 10 A particle P is moving in simple harmonic motion. The graph of **velocity** versus time is given below.



What is the amplitude of the motion?

- A $\frac{\pi}{12}$
 B $\frac{\pi}{6}$
 C $\frac{\pi}{3}$
 D $\frac{\pi}{2}$

SECTION 2 – 6 QUESTIONS EACH WORTH 15 MARKS**Question 11 – Begin a new writing booklet**

a Find $\int x\sqrt{1-x} \, dx$. **3**

b Find $\int_1^2 \frac{4}{x(x^2+4)} \, dx$. **3**

c Find $\int_0^1 \frac{dw}{1+\sqrt{w}}$. **3**

d The 4 complex numbers z_1, z_2, z_3, z_4 are represented by the points Z_1, Z_2, Z_3, Z_4 on an Argand diagram. $Z_1Z_2Z_3Z_4$ is a square.

i Prove that $|z_3 - z_1| - |z_4 - z_2| = 0$. **2**

ii Prove that $\arg\left(\frac{z_4 - z_2}{z_3 - z_1}\right) = \frac{\pi}{2}$. **2**

iii Prove that $z_3 - z_1 = (1+i)(z_2 - z_1)$. **2**

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Question 12 – Begin a new writing booklet

- a** Consider the statement:
 $\forall A \in \mathbb{R} : \text{If } \sin A = k \text{ then } \sin^{-1} k = A.$
- i** Write the converse. 2
- ii** Write the contrapositive. 2
- iii** Write the negation. 2
- iv** Determine whether or not the statement is an equivalence. Give reasons. 2
- v** Determine whether or not this statement is true. If true, justify using logic, or if not, give a counter-example. 2
- c** For the complex number $z = x + iy$ where $x, y \in \mathbb{R}$ sketch the set of z such that:
- i** $|z + 1 + i| = |z - 1 - i|$ 2
- ii** $|z + 1 + i| \leq 1$ 1
- iii** $z^4 = 16e^{i\pi}$. 2

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Question 13 – Begin a new writing booklet

a

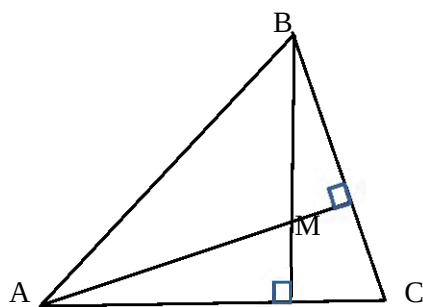
The two lines $\vec{r} = \begin{pmatrix} 1 \\ -3 \\ 3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 4 \\ -5 \\ 2 \end{pmatrix}$ and $\vec{q} = \begin{pmatrix} -1 \\ 6 \\ 7 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ intersect at (a, b, c) .

i Find (a, b, c) . **3**

ii Show that $\vec{r}$ and $\vec{q}$ are perpendicular. **2**

iii Find a vector which is perpendicular to both $\vec{r}$ and $\vec{q}$. **3**

b Consider the triangle ABC shown. The altitudes (perpendicular heights) from A to BC and B to AC intersect at M , i.e. $(\vec{a} - \vec{m}) \perp (\vec{b} - \vec{c})$, etc. **4**



Copy the diagram.

Use vectors to prove that CM is perpendicular to AB .

[This is the proof of the theorem that the altitudes of a triangle are concurrent.]

c Let $z = \cos \theta + i \sin \theta$. Using any method, factorise $z^6 + 1$ into 3 quadratic factors with real coefficients. **3**

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Question 14 – Begin a new writing booklet

- a i** If a, b are real, show that $a^2 + b^2 \geq 2ab$. **1**

Hence, or otherwise, show that if a, b, c are real then

- ii** $a^2 + b^2 + c^2 \geq ab + bc + ca$ **2**

- iii** $(a^2 + b^2 + c^2)^2 \geq 3abc(a + b + c)$. **2**

- b** Let $I_n = \int_0^1 x(1-x^3)^n dx$.

- i** Show that $I_n = \frac{3n}{3n+2} I_{n-1}$. **3**

- ii** Use the identity above to find I_3 . **2**

- c i** Show that $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$ for $|x| < 1$ and $|y| < 1$. **1**

- ii** Use mathematical induction to prove that **3**

$$\sum_{j=1}^n \tan^{-1} \left(\frac{1}{2j^2} \right) = \tan^{-1} \left(\frac{n}{n+1} \right).$$

- iii** Find $\lim_{n \rightarrow \infty} \sum_{j=1}^n \tan^{-1} \left(\frac{1}{2j^2} \right)$. **1**

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Question 15 – Begin a new writing booklet

- a** A suitcase of mass m kg on wheels is in equilibrium sitting on a flat rough surface. A person is trying to pull the suitcase at an angle of 60° to the horizontal with force S Newton. Gravity of g m/s² is acting downwards.
- i** Draw a diagram and resolve the forces in the horizontal and vertical directions. **2**
- ii** Find an expression for the coefficient μ of static frictional force F Newton on the object in terms of S , m and g . **2**
- b** A particle is moving horizontally at time t seconds, such that $v^2 = x(6 - x)$ where v is velocity in m/s and x is displacement in metres.
- i** Prove that it is moving in simple harmonic motion. **2**
- ii** Find the centre and amplitude. **2**
- iii** Find the period. **1**
- iv** Find the maximum speed. **1**
- v** At $t = 0$, $v = 3$ and $x = 3$. Find a formula for displacement x in terms of t . **2**
- vi** After 3 seconds, determine where the particle is, if the particle is travelling to the left or right, and if it is speeding up or slowing down. **3**

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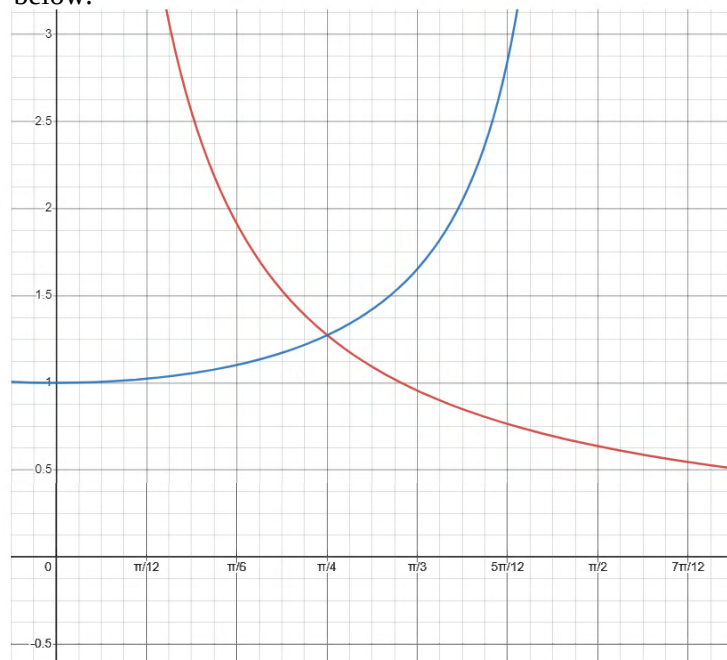
Question 16 – Begin a new writing booklet**a**

Use the substitution $u = \cos 2\theta$ to evaluate $\int_{\frac{1}{2}}^1 \sqrt{\frac{1-u}{1+u}} du$.

4**b**

Consider the graphs with equations $y = \frac{\tan x}{x}$ and $y = \frac{1}{x}$ for $x, y > 0$ shown

below.

**i**

Show that the curves intersect at $\left(\frac{\pi}{4}, \frac{4}{\pi}\right)$.

1**ii**

Show with justification that $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\tan x}{x} dx > \ln\left(\frac{4}{3}\right)$.

3

Question 16 continues on the next page...

Question 16 continued...

c A particle with acceleration $\ddot{x}$, displacement x and velocity v at time t moves such that $\ddot{x} = \frac{v}{50}(50 - v)$. Initially the particle is at the origin and the velocity is 100.

i Find an equation for v in terms of x . **3**

ii Find an equation for x in terms of t . **2**

iii Find the time taken for the particle to travel 25 units. **2**

The end! 😊

MC:

1. y-axis is $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$

$$\therefore \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 3 \times 0 + 0 \times 1 - 1 \times 0 = 0$$

$$\text{Test } \begin{pmatrix} 3 \\ 0 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = 3 \times 1 + 0 \times 2 - 1 \times 3 = 0$$

 $\therefore$ (A)

2. $|u| + |v| \geq |u+v|$ [Δ-ineq]

$$\therefore |2x-y| + |x+2y| \geq |2x-y+x+2y| \geq |3x+y|$$

(D)

3. $\frac{1}{x+iy} \times \frac{x-iy}{x-iy} = \frac{x-iy}{x^2+y^2} = a+ib$

$$\text{So } a = \frac{x}{x^2+y^2}, b = \frac{-iy}{x^2+y^2}$$

$$\therefore a, b \in \mathbb{Q} \text{ if } x, y \in \mathbb{N}$$

 $\therefore$ (A)

4. $1-2i$ is root so $1+2i$ is.
 $(1-2i)(1+2i) = 5$. Must be factors of 10 so other roots must be factors of 2.
 $(1+i)(1-i) = 2$ or 2 $\therefore$ (D)

5. $x=2, y=t, z=\cos t$
 $\therefore z = \cos y$ and x fixed.

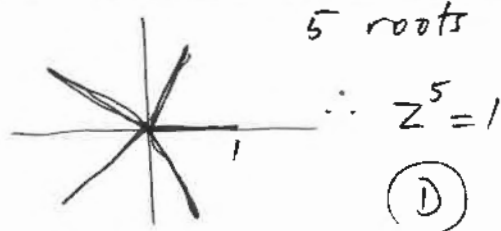
(C)

6. Contrapositive of $A \Rightarrow B$ is $\neg B \Rightarrow \neg A$.

$$\therefore \text{If } n < 3 \text{ then } x^n + y^n = z^n.$$

(D)

7.



5 roots

$$\therefore z^5 = 1$$

(D)

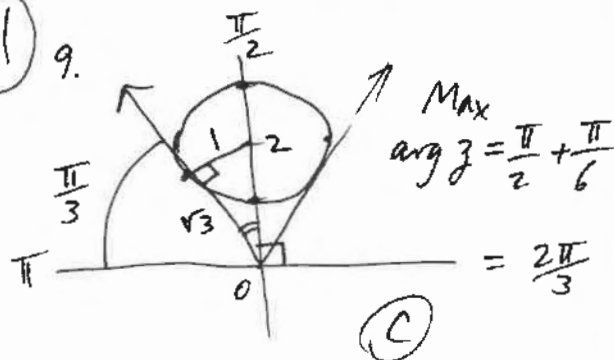
$$8. \sum_{i=1}^{99} i^n = i^1 + i^2 + i^3 + i^4 + i^5 + i^6 + \dots + i^{96} + i^{97} + i^{98} + i^{99}$$

$$= i^1 + i^2 + i^3 + i^4 + i^5 + i^6 + \dots + i^{96} + i^{97} + i^{98} + i^{99}$$

$$\dots + i^1 + i^2 + i^3 + i^4 + i^5 + i^6 + \dots + i^{96} + i^{97} + i^{98} + i^{99}$$

(B)

9.



(C)

$$10. \text{Period} = \frac{2\pi}{3} = \frac{2\pi}{n} \Rightarrow n=3.$$

$$\text{Max vel} = \frac{\pi}{2}.$$

$$\text{let } x = -A \cos(nt + \epsilon)$$

$$v = A n \sin(nt + \epsilon)$$

$$\therefore \text{Max when } \sin(nt + \epsilon) = 1$$

$$\therefore v = A n$$

$$\frac{\pi}{2} = A \times 3$$

$$\therefore A = \frac{\pi}{6}$$

(B)

SECTION 2

Q 11

$$a) \int x\sqrt{1-x} dx \quad \text{let } u=1-x$$

$$du = -dx$$

$$= \int (1-u)\sqrt{u} \cdot -du$$

$$= -\int u^{\frac{1}{2}} - u^{\frac{3}{2}} du$$

$$= -\left[\frac{2}{3}u^{\frac{3}{2}} - \frac{2}{5}u^{\frac{5}{2}}\right] + C$$

$$= \cancel{\frac{2}{3}(1-x)^{\frac{3}{2}} - \frac{2}{5}(1-x)^{\frac{5}{2}}} + C$$

$$= (1-x)\sqrt{1-x} \left[\frac{2}{5}(1-x) - \frac{2}{3}\right] + C$$

$$= (1-x)\sqrt{1-x} \left[-\frac{2}{5}x - \frac{4}{15}\right] + C$$

$$= (x-1)\sqrt{1-x} \left(\frac{2}{5}x + \frac{4}{15}\right) + C$$

$$b) \int_1^2 \frac{4}{x(x^2+4)} dx \equiv \int_1^2 \frac{A}{x} + \frac{Bx+C}{x^2+4} dx$$

$$\therefore A(x^2+4) + x(Bx+C) \equiv 4$$

$$B+A=0, \quad 4A=4, \quad C=0$$

$$\therefore B=-1, \quad A=1$$

$$\therefore \int_1^2 \frac{1}{x} + \frac{-x}{x^2+4} dx$$

$$= \left[\ln|x| - \frac{1}{2}\ln|x^2+4|\right]_1^2$$

$$= \ln 2 - \frac{1}{2}\ln 8 - \ln 1 + \frac{1}{2}\ln 5$$

$$= \ln 2 - \frac{1}{2}\ln 8 + \frac{1}{2}\ln 5$$

$$= \ln \left| \frac{x}{x\sqrt{2} \times \sqrt{5}} \right|$$

$$= -\frac{1}{2}\ln 10$$

$$c) \int_0^1 \frac{dw}{1+\sqrt{w}} \quad \text{let } u^2=w$$

$$2u du = dw$$

$$w=1 \quad u=1$$

$$w=0 \quad u=0$$

Q 11 c) Cont'd

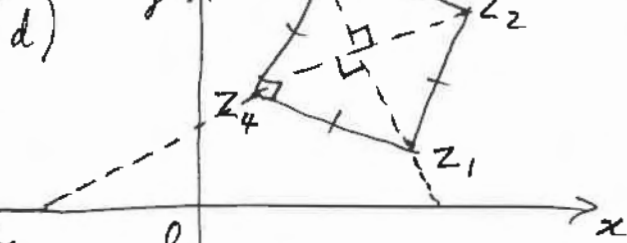
$$= 2 \int_0^1 \frac{u+1-1}{1+u} du$$

$$= 2 \int_0^1 1 - \frac{1}{1+u} du$$

$$= 2 \left[u - \ln|1+u| \right]_0^1$$

$$= 2 \left[1 - \ln 2 - 0 + \ln 1 \right]$$

$$= 2 - 2\ln 2$$



$$i) \text{ RTP: } |z_3 - z_1| - |z_4 - z_2| = 0$$

Proof: $z_1 z_3 = z_4 z_2$ since it is a square (diagonals equal)

$$\therefore |z_3 - z_1| = |z_4 - z_2|$$

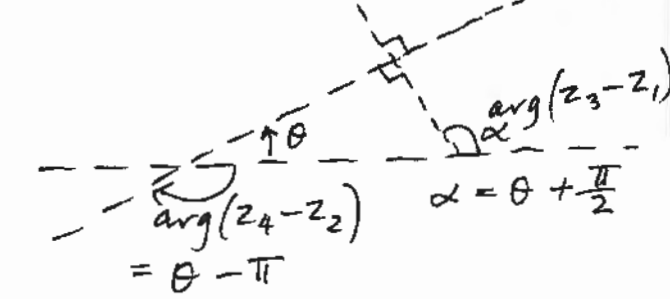
$$\therefore |z_3 - z_1| - |z_4 - z_2| = 0.$$

$$ii) \text{ RTP: } \arg\left(\frac{z_4 - z_2}{z_3 - z_1}\right) = \frac{\pi}{2}.$$

$$\text{Proof: } \arg\left(\frac{z_4 - z_2}{z_3 - z_1}\right)$$

$$= \arg(z_4 - z_2) - \arg(z_3 - z_1)$$

Close examination of the diagram:



Q 11 Cont'd ii) Cont'd

$$\therefore \arg(z_3 - z_1) = \alpha$$

$$\arg(z_4 - z_2) = \theta - \pi \text{ (negative)}$$

$$\therefore \arg(z_4 - z_2) - \arg(z_3 - z_1)$$

$$= \theta - \pi - \alpha \text{ but } \theta + \frac{\pi}{2} = \alpha$$

$$\therefore = \theta - \pi - (\theta + \frac{\pi}{2})$$

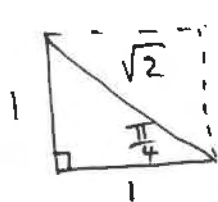
$$= \theta - \pi - \theta - \frac{\pi}{2}$$

$$= -\frac{3\pi}{2} = \frac{\pi}{2}$$

[Easier to say angle between 2 diagonals is 90° since square so $z_4 - z_2$ is $z_3 - z_1$ rotated 90° anticlockwise.]

$$\text{iii) RTP: } z_3 - z_1 = (1+i)(z_2 - z_1)$$

Proof: Multiply $z_2 - z_1$ by $1+i$ means rotating by $\frac{\pi}{4}$ ($\arg(1+i) = \frac{\pi}{4}$) and increasing by factor of $|1+i| = \sqrt{2}$.



Diagonal $z_3 - z_1$ is $|z_2 - z_1| \times \sqrt{2}$ and rotated 45° anticlockwise. (2)

iii) Negation: Not all A such that

$$\sin A = k \text{ means } \sin^{-1} k = A.$$

$$\text{or } \exists \sin A = k \text{ such that}$$

$$\sin^{-1} k \neq A. \quad (2)$$

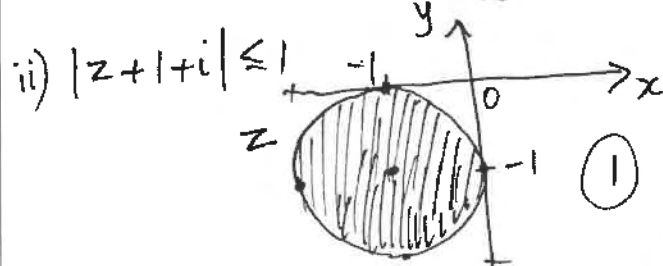
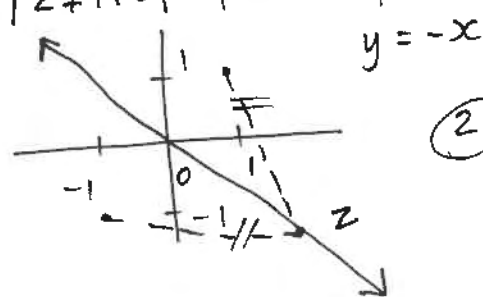
iv) Equivalence if $A \Rightarrow B$ and $B \Rightarrow A$ are both true. $A \Rightarrow B$ is

$$\text{NOT true (} \sin \frac{5\pi}{6} = \frac{1}{2} \text{ but } \sin^{-1} \frac{1}{2} = \frac{\pi}{6} \text{)}$$

but $B \Rightarrow A$ is true $\therefore$ Not an equivalence. (2)

v) Not true since contrapositive is not true. eg. $\sin^{-1} \frac{1}{\sqrt{2}} \neq \frac{3\pi}{4}$ but $\sin \frac{3\pi}{4} = \frac{1}{\sqrt{2}}$. FALSE (2) statement in the first place.

$$\text{c) i) } |z+1+i| = |z-1-i|$$



Q 12

a) $\forall A \in \mathbb{R}$: If $\sin A = k$ then $\sin^{-1} k = A$.

i) Converse: If $\sin^{-1} k = A$ then $\sin A = k$. (2)

ii) Contrapositive: If $\sin^{-1} k \neq A$ then $\sin A \neq k$. (2)

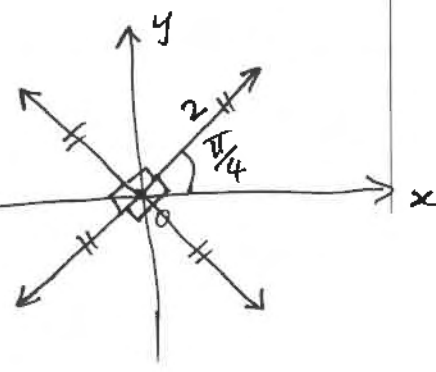
$$\text{iii) } z^4 = 16e^{i\pi}$$

$$z^4 = -16$$

$$4\cos 4\theta = -16$$

$$(2) \therefore r=2$$

$$\theta = \frac{\pi}{4}$$



$$Q13 a) \underline{r} = \begin{pmatrix} 1 \\ -3 \\ 3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 4 \\ -5 \\ 2 \end{pmatrix}$$

$$\underline{q} = \begin{pmatrix} -1 \\ 6 \\ 7 \end{pmatrix} + \lambda_2 \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}$$

$$\therefore i) 1 + 4\lambda_1 = -1 + \lambda_2 \Rightarrow \lambda_2 = 2 + 4\lambda_1$$

$$-3 - 5\lambda_1 = 6 + 2\lambda_2 \quad (2)$$

$$3 + 2\lambda_1 = 7 + 3\lambda_2 \quad (3)$$

Sub $\lambda_2 = 2 + 4\lambda_1$ into (2)

$$-3 - 5\lambda_1 = 6 + 2(2 + 4\lambda_1)$$

$$-3 - 5\lambda_1 = 6 + 4 + 8\lambda_1$$

$$-13 = 13\lambda_1 \Rightarrow \lambda_1 = -1$$

$$\therefore \lambda_2 = 2 + 4(-1) = -2$$

$$\therefore \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \\ 1 \end{pmatrix} \quad (3)$$

$$ii) \underline{r} \cdot \underline{q} = \begin{pmatrix} 4 \\ -5 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad (2)$$

$$= 4 \times 1 - 5 \times 2 + 2 \times 3$$

$$= 4 - 10 + 6$$

$$= 0 \therefore \underline{r} \perp \underline{q}$$

iii) Find $\begin{pmatrix} l \\ m \\ n \end{pmatrix}$ such that

$$\underline{r} \cdot \begin{pmatrix} l \\ m \\ n \end{pmatrix} = 0 \text{ and } \underline{q} \cdot \begin{pmatrix} l \\ m \\ n \end{pmatrix} = 0$$

$$\therefore \begin{pmatrix} 4 \\ -5 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} l \\ m \\ n \end{pmatrix} = 4l - 5m + 2n = 0 \quad (1)$$

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} l \\ m \\ n \end{pmatrix} = l + 2m + 3n = 0$$

$$\therefore 4l + 8m + 12n = 0 \quad (2)$$

$$(2) - (1)$$

$$13m + 10n = 0$$

$$\therefore 13m = -10n$$

$$m = \frac{-10n}{13}$$

Q13 cont'd: iii)

$$\therefore l + 2\left(\frac{-10n}{13}\right) + 3n = 0$$

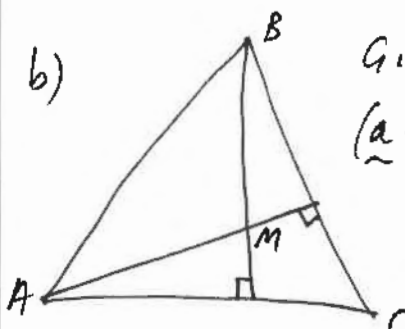
$$13l - 20n + 39n = 0$$

$$13l + 19n = 0 \quad (3)$$

$$l = \frac{-19n}{13}$$

$$\therefore \begin{pmatrix} l \\ m \\ n \end{pmatrix} = \begin{pmatrix} \frac{-19n}{13} \\ \frac{-10n}{13} \\ n \end{pmatrix} \text{ or } \begin{pmatrix} -19 \\ -10 \\ 13 \end{pmatrix}$$

$$\text{or } \begin{pmatrix} 19 \\ 10 \\ -13 \end{pmatrix}$$

b)  Given

$$(\underline{a} - \underline{m}) \cdot (\underline{b} - \underline{c}) = 0$$

(perpendicular)

$$\text{Also } (\underline{b} - \underline{m}) \cdot (\underline{a} - \underline{c}) = 0 \quad (\text{given})$$

RTP: $CM \perp AB$

$$\text{i.e. } (\underline{c} - \underline{m}) \cdot (\underline{a} - \underline{b}) = 0$$

$$\text{Proof: } (\underline{a} - \underline{m}) \cdot (\underline{b} - \underline{c}) = 0 \quad (1)$$

$$0 = \underline{a} \cdot \underline{b} - \underline{a} \cdot \underline{c} - \underline{m} \cdot \underline{b} + \underline{m} \cdot \underline{c}$$

$$\text{and } (\underline{b} - \underline{m}) \cdot (\underline{a} - \underline{c}) = 0 \quad (2)$$

$$\therefore 0 = \underline{b} \cdot \underline{a} - \underline{b} \cdot \underline{c} - \underline{m} \cdot \underline{a} + \underline{m} \cdot \underline{c}$$

~~Subtracting (1) & (2)~~ Subtracting:

$$0 = -\underline{b} \cdot \underline{c} + \underline{a} \cdot \underline{c} - \underline{m} \cdot \underline{a} + \underline{m} \cdot \underline{b}$$

$$\therefore 0 = \underline{c} \cdot (\underline{a} - \underline{b}) - \underline{m} \cdot (\underline{a} - \underline{b})$$

$$= (\underline{c} - \underline{m}) \cdot (\underline{a} - \underline{b}) \quad (4)$$

$\therefore CM \perp AB$ QED
[Altitudes all meet at M.]

Q13 cont'd:

c) $z = \cos \theta + i \sin \theta$

$$z^6 + 1 = (z^2 + 1)(z^4 - z^2 + 1)$$

$$= (z^2 + 1)(z^4 + 2z^2 + 1 - 3z^2)$$

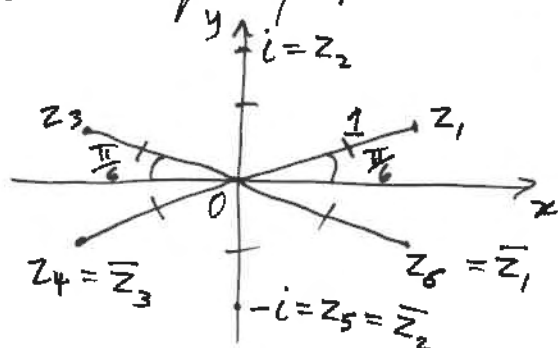
$$= (z^2 + 1)((z^2 + 1)^2 - 3z^2)$$

$$= (z^2 + 1)(z^2 + 1 - \sqrt{3}z)(z^2 + 1 + \sqrt{3}z)$$

$$= (z^2 + 1)(z^2 - \sqrt{3}z + 1)(z^2 + \sqrt{3}z + 1)$$

OR let $z^6 + 1 = 0 \therefore z^6 = -1 = \cos \pi$

Roots are $z_i = \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}$

then rest equally spaced $\frac{\pi}{3}$.

$$\therefore z^6 + 1 = (z - i)(z + i)(z - \cos \frac{\pi}{6} - i \sin \frac{\pi}{6})(z - \cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$$

$$\times (z - \cos \frac{5\pi}{6} - i \sin \frac{5\pi}{6})(z - \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6})$$

$$= (z^2 + 1)(z^2 - 2\cos \frac{\pi}{6}z + 1)(z^2 - 2\cos \frac{5\pi}{6}z + 1)$$

$$= (z^2 + 1)(z^2 - \sqrt{3}z + 1)(z^2 + \sqrt{3}z + 1)$$

[since $\cos(-\theta) = \cos \theta$] (3)

Q14 a) i) RTP: $a^2 + b^2 \geq 2ab$

Proof: Consider the difference:

$$a^2 + b^2 - 2ab = (a - b)^2$$

 ≥ 0 since square

$$\therefore a^2 + b^2 \geq 2ab. \quad (1)$$

ii) RTP: $a^2 + b^2 + c^2 \geq ab + ac + bc$

Proof: From (i) $a^2 + b^2 \geq 2ab$ (1)Similarly, $b^2 + c^2 \geq 2bc$ (2)Similarly, $c^2 + a^2 \geq 2ac$ (3)

Adding (1) + (2) + (3)

$$2a^2 + 2b^2 + 2c^2 \geq 2ab + 2bc + 2ac$$

$$\therefore a^2 + b^2 + c^2 \geq ab + bc + ca$$

iii) RTP: $(a^2 + b^2 + c^2)^2 \geq 3abc(a + b + c)$

Proof: Consider LHS:

$$(a^2 + b^2 + c^2)(a^2 + b^2 + c^2)$$

$$= a^4 + b^4 + c^4 + a^2b^2 + a^2c^2 + b^2a^2 + b^2c^2 + c^2a^2 + c^2b^2$$

$$= a^4 + b^4 + c^4 + 2(a^2b^2 + b^2c^2 + c^2a^2)$$

$$\geq a^4 + b^4 + c^4 + 2(abc^2 + bc^2a + ca^2b)$$

from (ii)

$$\geq a^4 + b^4 + c^4 + 2(abc(b + c + a))$$

$$\geq a^2b^2 + b^2c^2 + c^2a^2 + 2abc(b + c + a)$$

$$\geq abc(b + c + a) + 2abc(b + c + a)$$

$$\geq 3abc(a + b + c) \quad \text{QED} \quad (2)$$

Q14 cont'd

$$b) \text{ Let } I_n = \int_0^1 x(1-x^3)^n dx.$$

$$i) \text{ RTP: } I_n = \frac{3n}{3n+2} I_{n-1} \quad (3)$$

$$\begin{aligned} \text{Proof: } I_n &= \int_0^1 x(1-x^3)^n dx \\ &= \int_0^1 x(1-x^3)(1-x^3)^{n-1} dx \\ &= \int_0^1 x(1-x^3)^{n-1} dx - \int_0^1 x^4(1-x^3)^{n-1} dx \\ &= I_{n-1} + \frac{1}{3} \int_0^1 x^2 \cdot 3x^2(1-x^3)^{n-1} dx \end{aligned}$$

$$\begin{aligned} \text{let } u &= x^2 & dv &= -3x^2(1-x^3)^{n-1} dx \\ du &= 2x dx & v &= \frac{1}{n}(1-x^3)^n \end{aligned}$$

$$\begin{aligned} &= I_{n-1} + \frac{1}{3} [uv - \int v du] \\ &= I_{n-1} + \frac{1}{3} \left[\left[\frac{x^2}{n} (1-x^3)^n \right]_0^1 - \int_0^1 (1-x^3)^n 2x dx \right] \end{aligned}$$

$$I_n = I_{n-1} + \frac{1}{3} \left[\left[\frac{0-0}{n} \right] - \frac{2}{3n} I_n \right]$$

$$\therefore I_n + \frac{2}{3n} I_n = I_{n-1}$$

$$\frac{3n+2}{3n} I_n = I_{n-1}$$

$$\therefore I_n = \frac{3n}{3n+2} I_{n-1} \quad \text{QED!}$$

$$\begin{aligned} ii) I_3 &= \frac{3(3)}{3(3)+2} \times I_2 \\ &= \frac{9}{11} \left(\frac{3(2)}{3(2)+2} I_1 \right) \\ &= \frac{9}{11} \times \frac{6}{8} \left(\frac{3(1)}{3(1)+2} I_0 \right) \end{aligned}$$

14 b) ii) Cont'd

$$\begin{aligned} I_3 &= \frac{9}{11} \times \frac{6}{8} \times \frac{3}{5} \int_0^1 x(1-x^3)^0 dx \\ &= \frac{81}{220} \int_0^1 x dx \\ &= \frac{81}{220} \left[\frac{x^2}{2} \right]_0^1 \\ &= \frac{81}{220} \times \left(\frac{1}{2} - 0 \right) \\ &= \frac{81}{440} \end{aligned} \quad (2)$$

$$c) i) \text{ RTP: } \tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$$

$$\text{Proof: Consider let } \alpha = \tan^{-1} x \Rightarrow \tan \alpha = x$$

$$\beta = \tan^{-1} y \Rightarrow \tan \beta = y$$

$$\therefore \tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\therefore \tan(\tan^{-1} x + \tan^{-1} y) = \frac{x+y}{1-xy} \quad (1)$$

$$\therefore \tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x+y}{1-xy}$$

[$\tan^{-1} x$ is a one-to-one fn.]

$$ii) \text{ RTP: Let } P(n) \text{ be the proposition that}$$

$$\sum_{j=1}^n \tan^{-1} \left(\frac{1}{2j^2} \right) = \tan^{-1} \left(\frac{n}{n+1} \right).$$

Proof: See over page!

Proof: RTP:

$$\tan^{-1}\left(\frac{1}{2(1)^2}\right) + \tan^{-1}\left(\frac{1}{2(2)^2}\right) + \tan^{-1}\left(\frac{1}{2(3)^2}\right) + \dots + \tan^{-1}\left(\frac{1}{2(n)^2}\right) = \tan^{-1}\left(\frac{n}{n+1}\right).$$

Prove $P(1)$ true:

$$\text{LHS} = \tan^{-1}\left(\frac{1}{2}\right) \quad \text{RHS} = \tan^{-1}\left(\frac{1}{1+1}\right) = \tan^{-1}\left(\frac{1}{2}\right)$$

$$\therefore P(1) \text{ true.} \quad = \text{LHS}$$

Assume $P(k)$ true i.e.

$$\tan^{-1}\left(\frac{1}{2(1)^2}\right) + \tan^{-1}\left(\frac{1}{2(2)^2}\right) + \tan^{-1}\left(\frac{1}{2(3)^2}\right) + \dots + \tan^{-1}\left(\frac{1}{2(k)^2}\right) = \tan^{-1}\left(\frac{k}{k+1}\right)$$

for some $k \in \mathbb{N}$.RTP: $P(k+1)$ true: i.e.

$$\tan^{-1}\left(\frac{1}{2(1)^2}\right) + \dots + \tan^{-1}\left(\frac{1}{2(k)^2}\right) + \tan^{-1}\left(\frac{1}{2(k+1)^2}\right) = \tan^{-1}\left(\frac{k+1}{k+2}\right) \therefore P(k+1) \text{ is true}$$

 $\therefore P(n)$ true by Math Induction.Proof: Consider the LHS of $P(k+1)$

$$\therefore \tan^{-1}\left(\frac{1}{2(1)^2}\right) + \dots + \tan^{-1}\left(\frac{1}{2(k)^2}\right) + \tan^{-1}\left(\frac{1}{2(k+1)^2}\right)$$

$$= \tan^{-1}\left(\frac{k}{k+1}\right) + \tan^{-1}\left(\frac{1}{2(k+1)^2}\right) \text{ using } P(k)$$

$$= \tan^{-1}\left(\frac{\frac{k}{k+1} + \frac{1}{2(k+1)^2}}{1 - \left(\frac{k}{k+1} \cdot \frac{1}{2(k+1)^2}\right)}\right) \quad |x| < 1, |y| < 1$$

$$= \tan^{-1}\left(\frac{2(k+1)k + 1}{2(k+1)^3 - k}\right)$$

Q14 Cont'd

c) ii) Cont'd:

$$\begin{aligned} &= \tan^{-1}\left(\frac{2k^2 + 2k + 1}{2(k+1)^3 - k}\right) \\ &= \tan^{-1}\left(\frac{2k^2 + 2k + 1}{2(k^3 + 3k^2 + 3k + 1) - k}\right) \\ &= \tan^{-1}\left(\frac{2k^2 + 2k + 1}{2k^3 + 6k^2 + 6k + 2 - k}\right) \\ &= \tan^{-1}\left(\frac{2k^2 + 2k + 1}{2k^3 + 6k^2 + 5k + 2}\right) \\ &= \tan^{-1}\left(\frac{(2k^2 + 2k + 1)(k+1)}{(2k^2 + 2k + 1)(k+2)}\right) \\ &= \tan^{-1}\left(\frac{k+1}{k+2}\right) \end{aligned}$$

check this step!

 $= \text{RHS of } P(k+1)$

③

$$\text{iii) } \lim_{n \rightarrow \infty} \sum_{j=1}^n \tan^{-1}\left(\frac{1}{2j^2}\right)$$

$$= \lim_{n \rightarrow \infty} \tan^{-1}\left(\frac{n}{n+1}\right)$$

$$= \lim_{n \rightarrow \infty} \tan^{-1}\left(\frac{n}{n}\right)$$

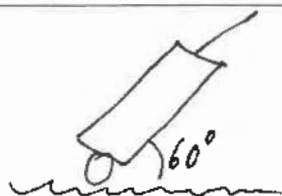
$$\approx \lim_{n \rightarrow \infty} \tan^{-1} 1$$

$$= \frac{\pi}{4}$$

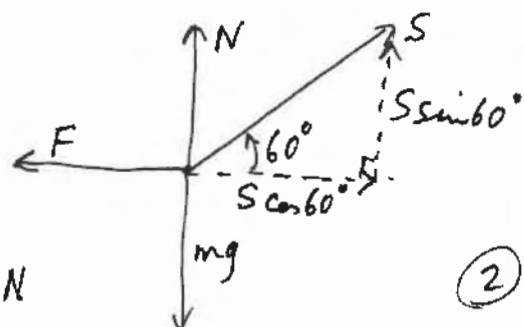
①

Q15

a)



i)



$$F = \mu N$$

(2)

Horizontally: $S \cos 60^\circ - F = 0$
 $F = S \cos 60^\circ = \frac{1}{2} S$

Vertically: $N + S \sin 60^\circ - mg = 0$
 $N = mg - \frac{\sqrt{3}}{2} S$

ii) $\mu = ?$ S, m, g . $\mu = \frac{F}{N}$

$$\mu = \frac{\frac{1}{2} S}{mg - \frac{\sqrt{3}}{2} S}$$

(2)

$$\therefore \mu = \frac{S}{2mg - \sqrt{3} S}$$

b) $v^2 = x(6-x) = 6x - x^2$

i) RTP: form is $\ddot{x} = -n^2(x-c)$

$$\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$$

(2)

$$= \frac{d}{dx} \left(\frac{1}{2} (6x - x^2) \right)$$

$$= 3 - x$$

$$= -1^2(x-3) \text{ which is}$$

$$\text{SHM } c=3, n=1$$

15 cont'd

b) ii) Centre is $x=3$. (2)

End points when $v=0$

$$0 = 6x - x^2 = x(6-x)$$

$$x=0 \text{ or } x=6, \text{ centre } 3$$

 $\therefore$ Amplitude is 3 units.

iii) Period $= \frac{2\pi}{n} = \frac{2\pi}{1} = 2\pi$. (1)

iv) Max speed occurs at centre
 when $x=3$ $|v| = \sqrt{3(6-3)}$
 $= 3$. (1)

v) $t=0, v=3, x=3$.

$$\therefore v = +\sqrt{x(6-x)}$$

$$\therefore \frac{dx}{dt} = \sqrt{6x - x^2}$$

$$\therefore \int_3^x \frac{1}{\sqrt{6x - x^2}} dx = \int_0^t dt$$

$$\int_3^x \frac{1}{\sqrt{9 - (x-3)^2}} dx = t$$

(2)

$$\left[\sin^{-1} \left(\frac{x-3}{3} \right) \right]_3^x = t$$

$$\sin^{-1} \frac{x-3}{3} - \sin^{-1} 0 = t$$

$$\therefore \frac{x-3}{3} = \sin t$$

$$x = 3 \sin t + 3$$

vi) $t=3, x = 3 \sin 3 + 3 \doteq 0.423360...$
 to Right

(3) $v = 3 \cos 3 = -2.9699...$ to left

$$\ddot{x} = -3 \sin 3 = -0.4233...$$

$\therefore$ P is ≈ 0.423 to right of 0,
 speeding up, going left.

Q16 a) $u = \cos 2\theta$

$$du = -2 \sin 2\theta d\theta$$

$$u = 1, \theta = \pi \text{ or } 0$$

$$u = \frac{1}{2}, \theta = \frac{\pi}{6}$$

$$\therefore \int_{\frac{1}{2}}^1 \sqrt{\frac{1-u}{1+u}} du$$

$$= \int_{\frac{\pi}{6}}^0 \sqrt{\frac{1-\cos 2\theta}{1+\cos 2\theta}} \times -2 \sin 2\theta d\theta$$

$$= \int_0^{\frac{\pi}{6}} \sqrt{\frac{2 \sin^2 \theta}{2 \cos^2 \theta}} \times 2 \sin \theta \cos \theta d\theta$$

$$= \int_0^{\frac{\pi}{6}} \sqrt{\frac{\sin^2 \theta}{\cos^2 \theta}} \cdot 2 \sin \theta \cos \theta d\theta$$

$$= \int_0^{\frac{\pi}{6}} \frac{\sin \theta}{\cos \theta} \cdot 2 \sin \theta \cos \theta d\theta$$

$$= \int_0^{\frac{\pi}{6}} 2 \sin^2 \theta d\theta \quad (4)$$

$$= \int_0^{\frac{\pi}{6}} 1 - \cos 2\theta d\theta$$

$$= \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{6}}$$

$$= \frac{\pi}{6} - \frac{1}{2} \sin \frac{\pi}{3} - \left(0 - \frac{1}{2} \sin 0 \right)$$

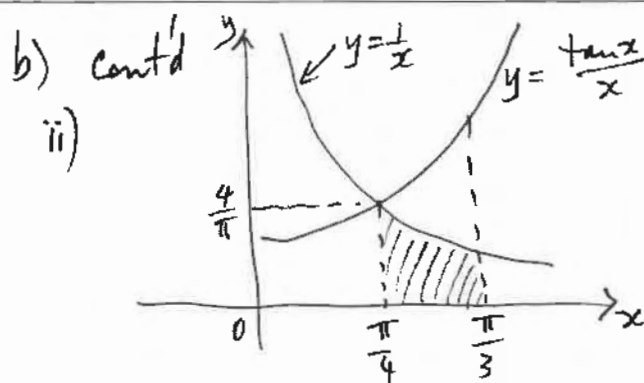
$$= \frac{\pi}{6} - \frac{\sqrt{3}}{4}$$

b) $y = \frac{\tan x}{x}, y = \frac{1}{x}$

i) Sub $\left(\frac{\pi}{4}, \frac{4}{\pi} \right)$: $y = \frac{4}{\pi}$, RHS = $\frac{\tan \frac{\pi}{4}}{\frac{\pi}{4}} = \frac{4}{\pi}$

$$y = \frac{1}{\frac{\pi}{4}} = \frac{4}{\pi} \quad (1)$$

$\therefore$ RHS = LHS for both
 $\therefore$ Intersects there.



Clearly from diagram,
Area below $y = \frac{\tan x}{x} >$ Area below $y = \frac{1}{x}$

between $x = \frac{\pi}{4}$ and $\frac{\pi}{3}$.

$$\therefore \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\tan x}{x} dx > \int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{1}{x} dx$$

$$> \left[\ln x \right]_{\frac{\pi}{4}}^{\frac{\pi}{3}}$$

$$> \ln \frac{\pi}{3} - \ln \frac{\pi}{4}$$

$$> \ln \left(\frac{\frac{\pi}{3}}{\frac{\pi}{4}} \right)$$

$$> \ln \frac{4}{3} \quad \text{QED!}$$

c) $\ddot{x} = \frac{v}{50} (50 - v)$

$$t = 0, x = 0, v = 100$$

i) let $\frac{v dv}{dx} = \frac{v}{50} (50 - v)$

$$\int \frac{v dv}{50 - v} = \frac{1}{50} \int dx$$

$$\left[-\ln |50 - v| \right]_{100}^v = \frac{1}{50} x$$

P.T.O.

Q16 cont'd

c) i) cont'd

$$\therefore -[\ln|50-v| - \ln|50-100|] = \frac{1}{50}x$$

$$\therefore -\left[\ln\left|\frac{50-v}{-50}\right|\right] = \frac{1}{50}x$$

$$\ln\left|\frac{-50}{50-v}\right| = \frac{1}{50}x \quad (3)$$

$$\frac{-50}{50-v} = e^{\frac{1}{50}x}$$

$$-50e^{-\frac{1}{50}x} = 50-v$$

$$\therefore v = 50 + 50e^{-\frac{1}{50}x}$$

$$\therefore v = 50(1 + e^{-\frac{1}{50}x})$$

ii) $\frac{dx}{dt} = 50(1 + e^{-\frac{1}{50}x})$

$$\int_0^x \frac{dx}{1 + e^{-\frac{1}{50}x}} = \int_0^t 50 dt \quad (2)$$

$$\int_0^x \frac{e^{\frac{1}{50}x}}{e^{\frac{1}{50}x} + 1} dx = 50t$$

$$\left[\ln|e^{\frac{1}{50}x} + 1|\right]_0^x = 50t$$

$$\ln|e^{\frac{1}{50}x} + 1| - \ln|e^0 + 1| = 50t$$

$$\ln\left|\frac{e^{\frac{1}{50}x} + 1}{2}\right| = 50t$$

$$\frac{e^{\frac{1}{50}x} + 1}{2} = e^{50t}$$

$$e^{\frac{1}{50}x} + 1 = 2e^{50t}$$

16 c) ii) cont'd

$$e^{\frac{1}{50}x} = 2e^{50t} - 1$$

$$\therefore \frac{1}{50}x = \ln|2e^{50t} - 1|$$

$$x = 50 \ln|2e^{50t} - 1|$$

iii) $x = 25, t = ?$

$$25 = 50 \ln|2e^{50t} - 1|$$

$$\frac{1}{2} = \ln|2e^{50t} - 1|$$

$$e^{\frac{1}{2}} = 2e^{50t} - 1 \quad (2)$$

$$\therefore e^{50t} = \frac{e^{\frac{1}{2}} + 1}{2}$$

$$\therefore 50t = \ln\left|\frac{\sqrt{e} + 1}{2}\right|$$

$$t = \frac{1}{50} \ln\left|\frac{\sqrt{e} + 1}{2}\right|$$

$$\approx 0.005618596... \text{ units!}$$

ASCHAM SCHOOL

YEAR 12 Trial Mathematics Extension 2 Exam

MULTIPLE-CHOICE ANSWER SHEET

- | | | | | |
|-----|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1. | A ☒ | B ☐ | C ☐ | D ☐ |
| 2. | A ☐ | B ☐ | C ☐ | D ☒ |
| 3. | A ☒ | B ☐ | C ☐ | D ☒ |
| 4. | A ☐ | B ☐ | C ☐ | D ☒ |
| 5. | A ☐ | B ☐ | C ☒ | D ☐ |
| 6. | A ☐ | B ☐ | C ☐ | D ☒ |
| 7. | A ☐ | B ☐ | C ☐ | D ☒ |
| 8. | A ☐ | B ☒ | C ☐ | D ☐ |
| 9. | A ☐ | B ☐ | C ☒ | D ☐ |
| 10. | A ☐ | B ☒ | C ☐ | D ☐ |